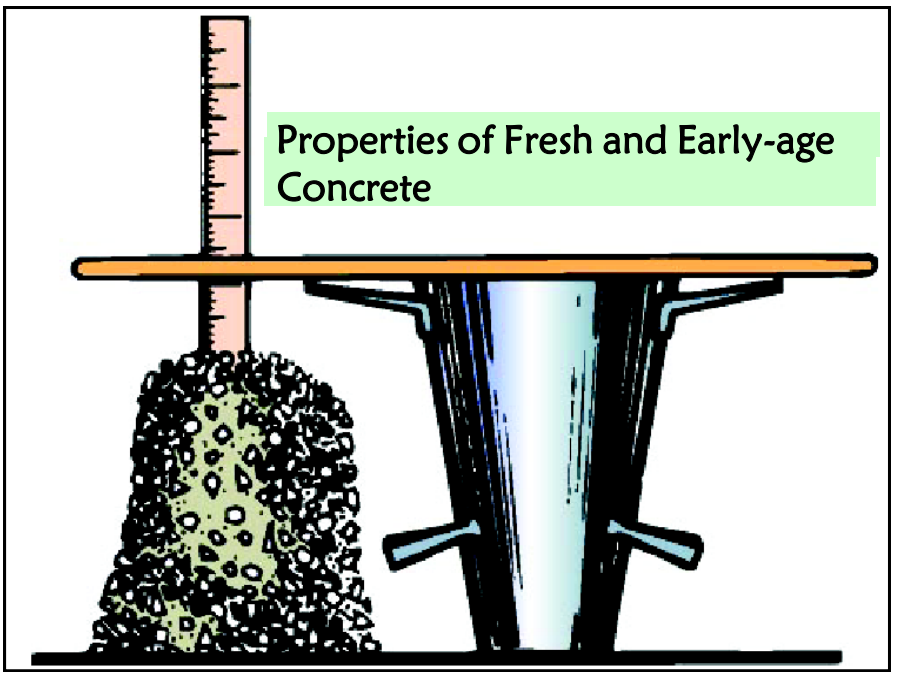
Building Materials

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Email: yawei@tsinghua.edu.cn

Chapter One Introduction
Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
Chapter Five Asphalt
Chapter Six Masonry
Chapter Seven Wood



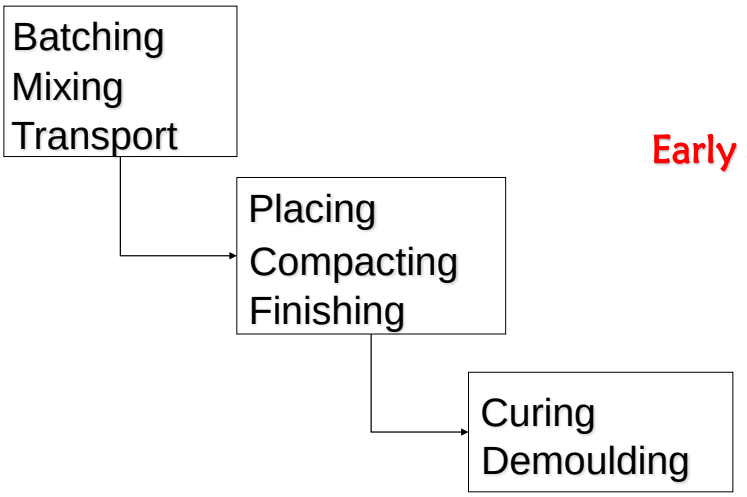
4.6 Fresh and Early-age Properties

Why we should care about early-age concrete?

Concrete resembles a human child: to develop into a healthy person, a newborn baby needs special attention during the early periods of growth. Deficiencies in fresh concretes can lead to serious material quality problems.

Mehta and Monteiro 2006

4.6 Fresh and Early-age Properties



4.6 Fresh and Early-age Properties

- Workability
- Segregation
- Bleeding
- Setting Time
- Uniformity

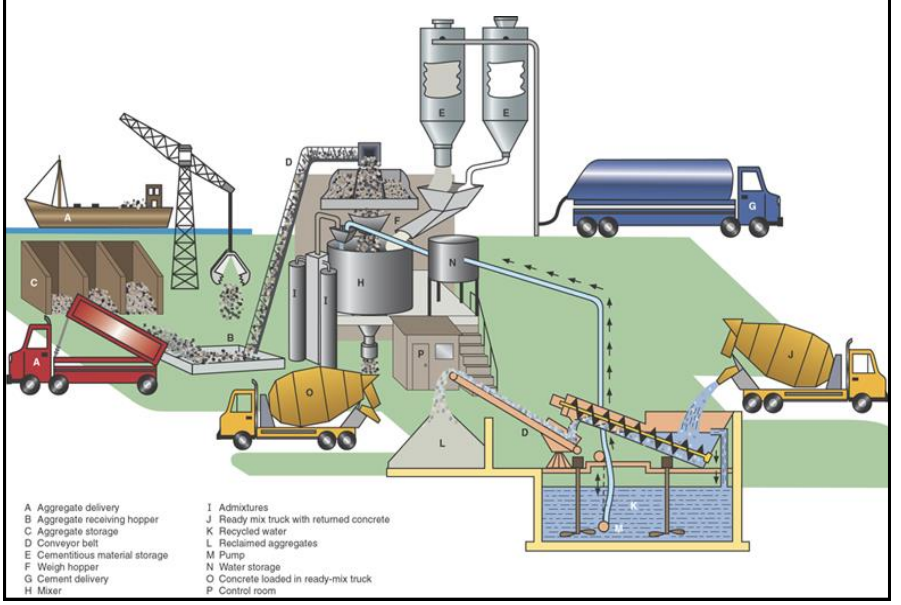
4.6 Fresh and early-age Properties

Mixing

The aim of mixing is to blend all of the ingredients of the concrete to form a uniform mass and to coat the surface of aggregates with cement paste.

- Ready-Mix concrete: In this type ingredients are introduced into a mixer truck and mixed during transportation to the site.
  - Wet – Water added before transportation
  - Dry – Water added at site
- Mixing at the site
  - Hand mixed
  - Mixer mixed

Ready Mix Concrete



4.6 Fresh and early-age Properties

Mixing

Mixing at Site





4.6 Fresh and early-age Properties

Mixing

Mixing time should be sufficient to produce a uniform concrete. The time of mixing depends on the type of mixer and also to some properties of fresh concrete.

- Undermixing → non-homogeneity
- Overmixing → danger of water loss, brekage of aggregate particles

4.6 Fresh and Early-age Properties

Workability

- Workability is the most important property of freshly mixed concrete.
- There is no single test method that can simultaneously measure all the properties involved in workability.



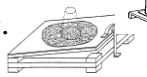
4.6 Fresh and Early-age Properties

Measurements of Workability

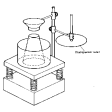
There are 5 types of test, which can measure workability indirectly. Unfortunately, there are no accepted tests which can measure directly the workability.







- Slump Test: Gives good results for rich mixes.
- Compacting Factor Test: Used for low workable concretes.
- Flow Table Test: Used for high workable concretes.
- VeBe Test: Used for low workable concretes.
- Kelly Ball Test: It is practical in field test.

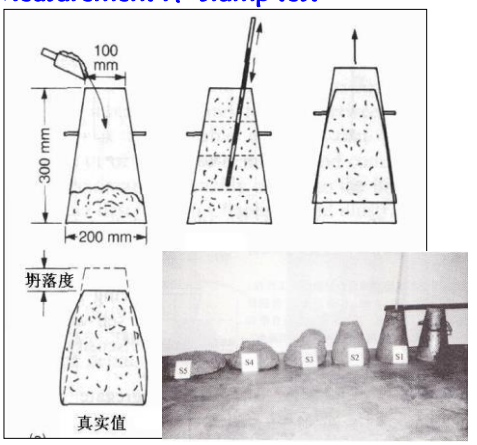




4.6 Fresh and Early-age Properties

Workability

Measurement 1： Slump test



This is a test used extensively in site work all over the world. Very useful in detecting variations in the uniformity of a mix of given nominal proportions.

4.6 Fresh and Early-age Properties

Workability

Measurement 1： Slump test

Slump Test is related with the ease with which concrete flows during placement.





4.6 Fresh and Early-age Properties

Workability

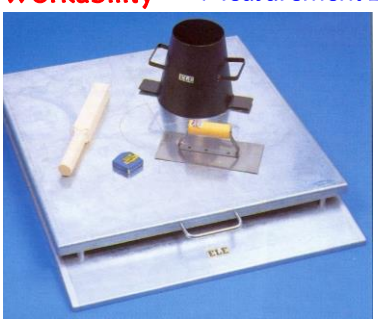
Measurement 1： Slump test

Recall …

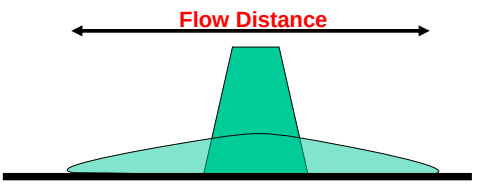
Types of Construction	Slump (in.) Maximum*	Minimum
Reinforced foundation walls and footings	3	1
Plain footings, caissons, and substructure walls	3	1
Beams and reinforced walls	4	1
Building columns	4	1
Pavements and slabs	3	1
Mass concrete	2	1

4.6 Fresh and Early-age Properties

Workability **Measurement 2： Flow Table test**



- Flow table test measures the horizontal spread of a concrete cone specimen.



- The test is especially appropriate for highly fluid mixes.
- The test apparatus is large and must be placed on firm, level ground.
- The further the concrete spreads, the thinner the layer of concrete becomes and the less this thin layer represents the bulk properties of the concrete.

4.6 Fresh and Early-age Properties

Bleeding and Segregation

- Segregation is separation of the constituents of a heterogeneous mixture so that their distribution is no longer uniform.

There are two forms of segregation:

- first form: coarse particles tend to separate out since they tend to settle more than fine particles.
- second form: occurs in wet mixes; it is manifested by the separation of (cement+water) from the mix.





4.6 Fresh and Early-age Properties

Bleeding and Segregation

- Bleeding (water gain) is a form of segregation, in which some of the water in the mix tends to rise to the surface of freshly placed concrete.
- Result: Top of every lift may become too wet and if the water is trapped by concrete, porous, weak, and non-durable concrete will result.
- If the bleeding water is remixed during finishing of the top surface a weak wearing surface will be formed.
- This can be avoided by delaying the finishing operations until the Bleeding water has evaporated.
- Some of the rising water becomes trapped on the underside of coarse aggregate particles or of reinforcement, thus creating zones of poor bond.

4.6 Fresh and Early-age Properties

Bleeding and Segregation

Is it always harmful?

Bleeding need not necessarily be harmful. If it is undisturbed and the water evaporates the effective water/cement ratio may be lowered with a resulting increase in strength.

If the rising water carries with it a considerable amount of the finer cement particles a layer of scum will be formed. Scum should always be removed by brushing and washing.

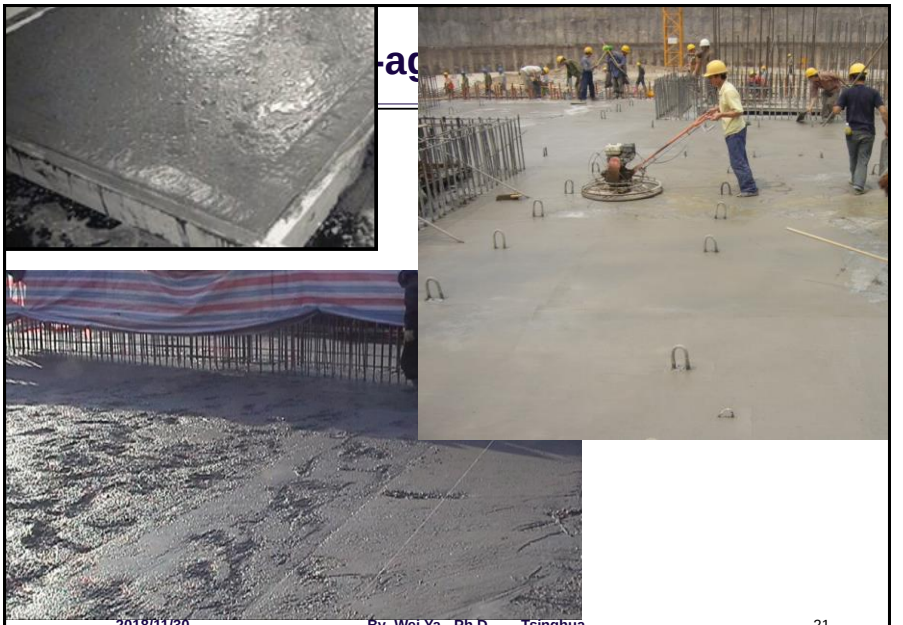
How can we reduce bleeding?

Bleeding is decreased by increasing the fineness of cement.

Rich mixes are less prone to bleeding than lean ones.

Reduction in bleeding is obtained by the addition of pozzolans.

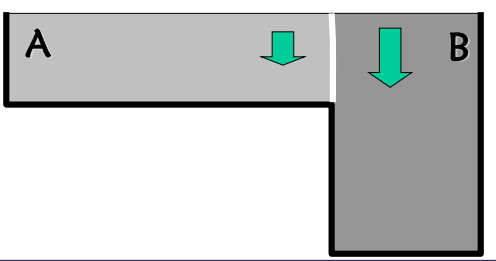
Air entrainment effectively reduces bleeding so that finishing can follow casting without delay.



4.6 Fresh and Early-age Properties

Subsidence

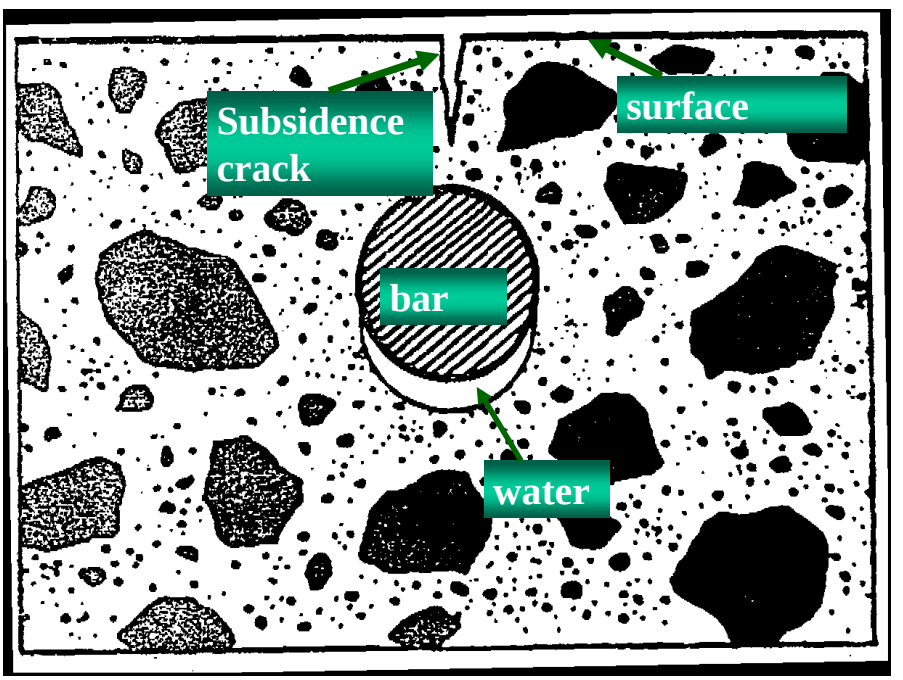
- Subsidence refers to the vertical shrinkage of fresh cementitious materials before initial set.
- It is caused by bleeding (settlement of solids relative to liquids), air voids rising to the surface, and chemical shrinkage.



4.6 Fresh and Early-age Properties

Subsidence

- Excessive subsidence is often caused by a lack of consolidation of fresh concrete. Excessive subsidence over embedded items, such as supported steel reinforcement, can result in cracking over embedded items.
- Concretes made with air entrainment, sufficient fine materials, and low water contents will minimize subsidence cracking. Also, plastic fibers have been reported to reduce subsidence cracking

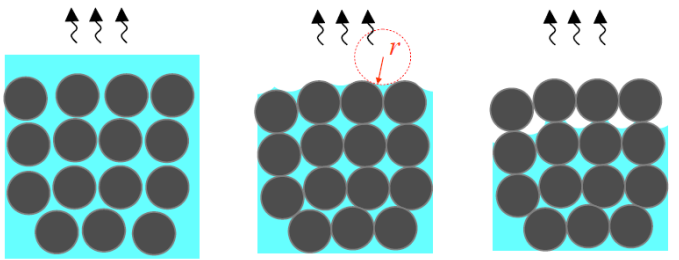


4.6 Fresh and Early-age Properties

Plastic shrinkage

Fast evaporation due to:

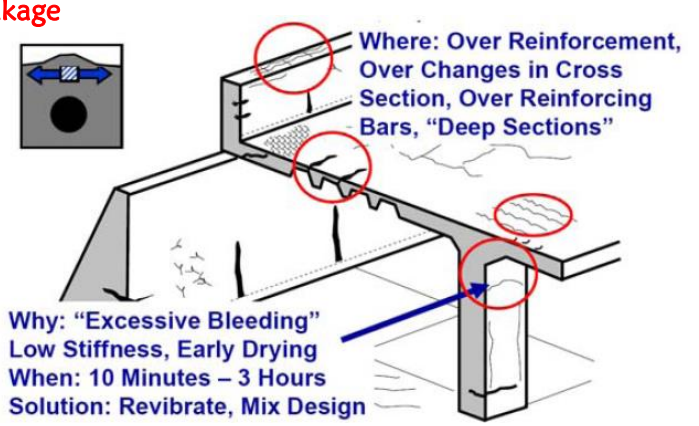
- Low relative humidity
- High wind speed
- High temperature, surface exposed to sun





4.6 Fresh and Early-age Properties

Plastic shrinkage



Differential settlement causes stress concentrations and cracks may be a problem in very fluid mixtures

4.6 Fresh and Early-age Properties

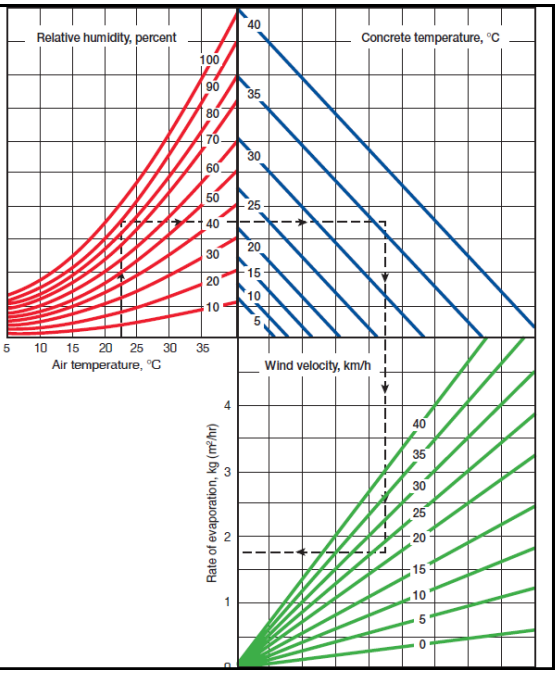
Plastic shrinkage

Plastic shrinkage cracks sometimes occur in the surface of freshly mixed concrete soon after it has been placed, while it is being finished or shortly thereafter. These cracks which appear mostly on horizontal surfaces can be substantially eliminated if preventive measures are taken.





4.6 Fresh and Early-age



Worst case:

- High evaporation rate
-Slow concrete hardening

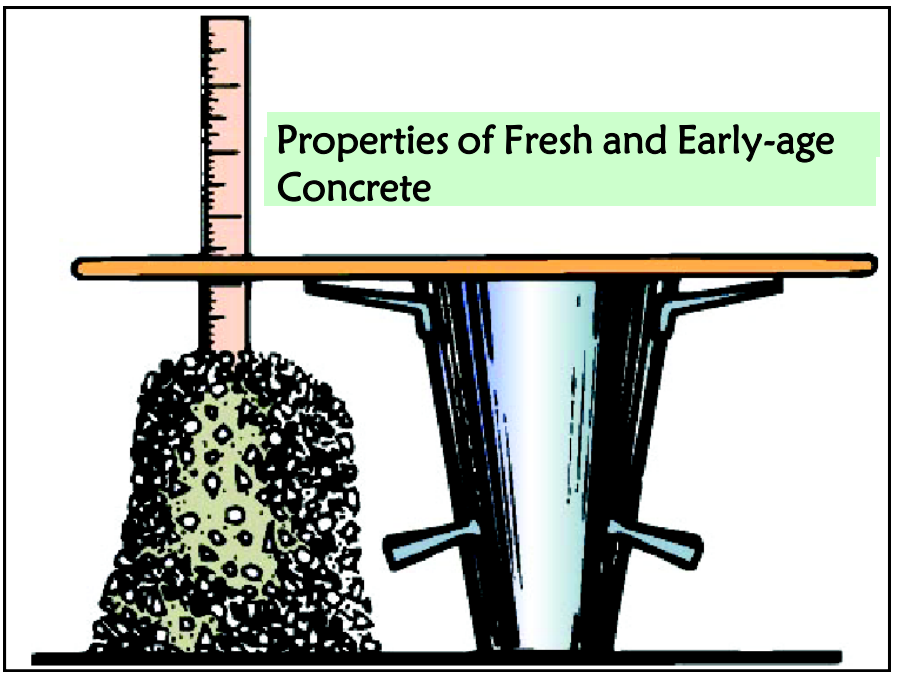
4.6 Fresh and Early-age Properties

Plastic shrinkage



Schmidt et al. BUST 2007

Sunny and windy day, slab 25 cm thick. Cracking within the first 4 hours after placing the concrete, crack distance ~1 m

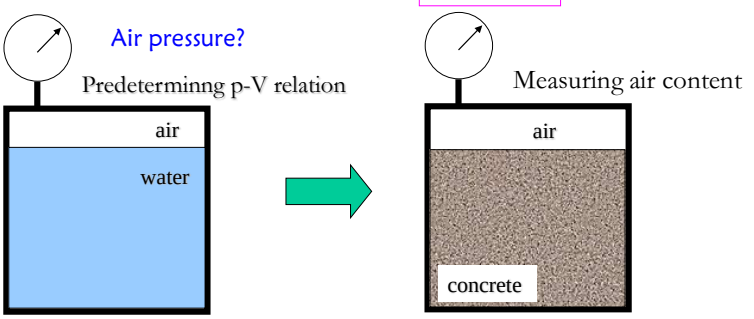


4.6 Fresh and Early-age Properties

Air Content Pressure Method For Air Content

Boyle's Law:

- At a constant temperature, the volume of a gas is inversely proportional to its pressure.
- Pressure × Volume at one time = Pressure × Volume at other times, if temperature does not change. $p_1V_1 = p_2V_2$



4.6 Fresh and Early-age Properties

Air Content Pressure Method For Air Content

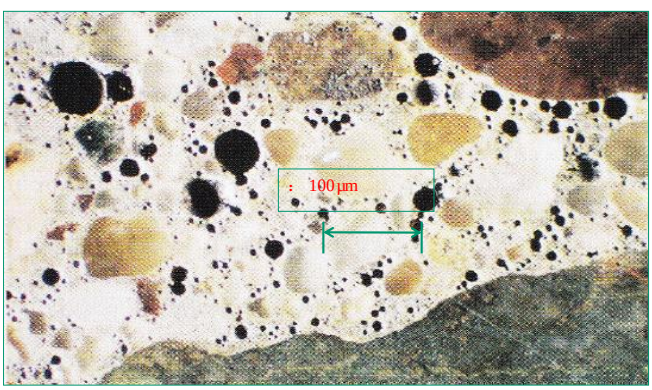
- The instrument should be calibrated for various elevations above sea level if it is to be used in locations having considerable differences in elevation.

Pressure-type meter for determining air content



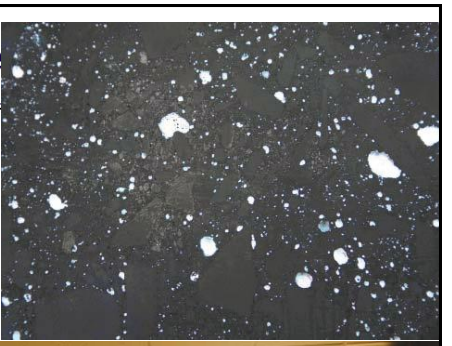
4.6 Fresh and Early-age Properties

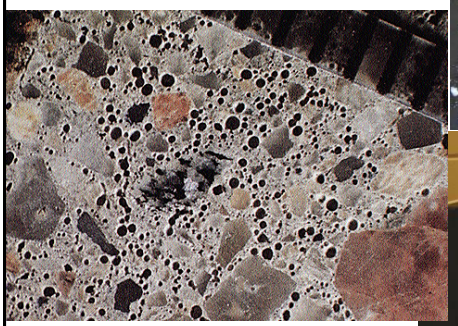
Air Content



4.6 Fresh and Early-age

Air Content





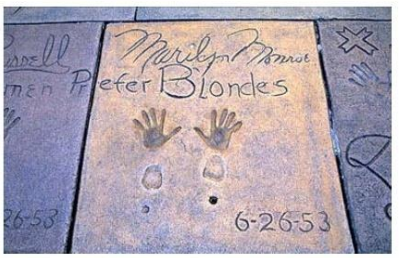


4.6 Fresh and Early-age Properties

Setting

Hardening of concrete

Plastic phase ⇒ Setting ⇒ Hard concrete



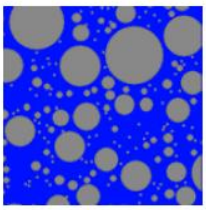


Hollywood, California

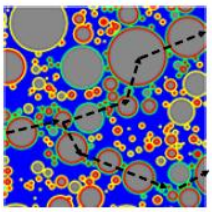
4.6 Fresh and Early-age Properties

Setting

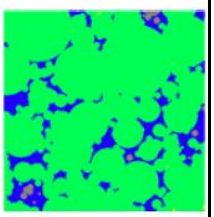
Setting and hardening process



Suspension



Threshold of solid percolation



Solids fully connected

Ultra sound

4.6 Fresh and Early-age Properties

Setting time：the time from the addition of water to the initial & final setting stage. Also refers to time of changes of the cement paste from a liquid to a rigid stage.

Initial Setting: The time at which the concrete can no longer be properly mixed, finished or compacted. (Represented by a Vicat needle penetration of 25 mm or less). It corresponds to a rapid rise in temperature. Normally takes about 45 – 175 minutes.

Final Setting: The time required for the cement to harden to a point where it can sustain some load. (Represented by no penetration of Vicat needle.), corresponds to the peak temperature in the cement paste. Normally takes between 3 hours to 10 hours for this to happen.

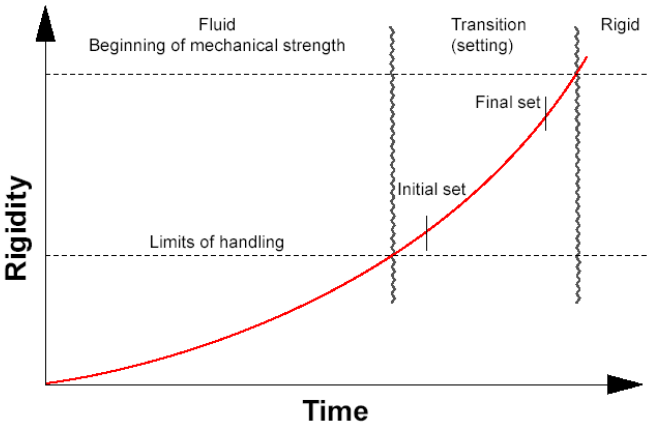
Hardening:

- Referred to the gained of the strength of the cement paste.
- During the setting time the cement gained very little strength.

Hardening is the development of strength over an extended period of time, is not completed for months or years.

4.6 Fresh and Early-age Properties

Setting

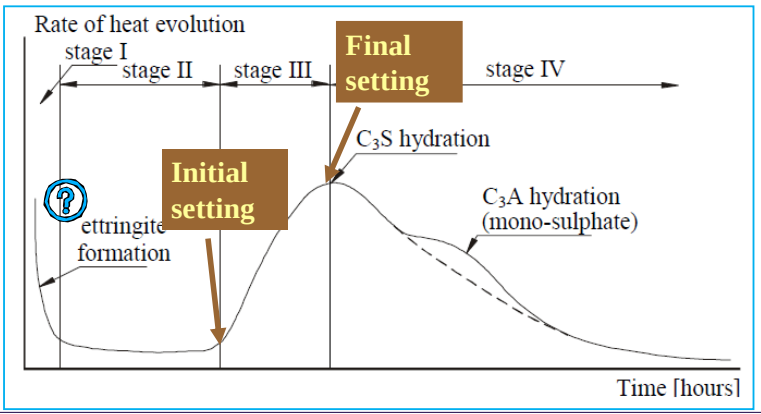


From Mindess and Young 1981

4.6 Fresh and Early-age Properties

Setting

Setting time is affected by: cement composition, cement fineness, rate of hydration, and the ambient temperature.



4.6 Fresh and Early-age Properties

Setting Cement Paste





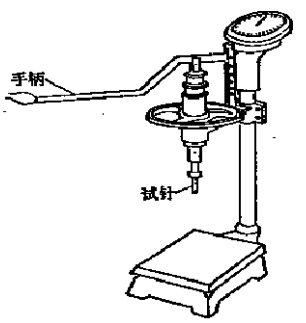
- ✓ The time taken for a 1-mm diameter needle in the Vicat apparatus to penetrate a depth of 25mm into the cement past sample is the initial setting time.

4.6 Fresh and Early-age Properties

Setting Concrete

Setting times of concretes do not correlate directly with setting times of pastes because of water loss to the air or substrate, presence of aggregate, and because of temperature differences in the field.

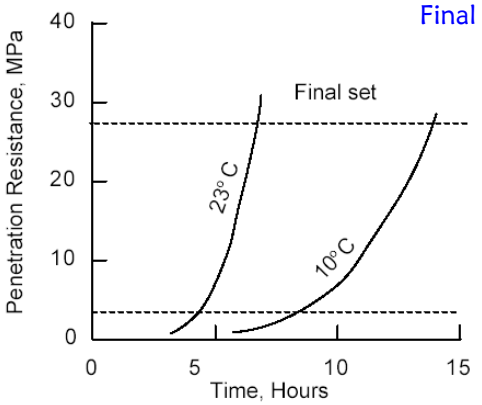
Initial set: 3.5 Mpa
Final set: 27.6 MPa



4.6 Fresh and Early-age Properties

Setting

Initial set: 3.5 Mpa
Final set: 27.6 MPa



From Sprouse and Pepler 1978

4.6 Fresh and Early-age Properties

Case study

On 27 April 1978 a cooling tower under construction at Willow Island in West Virginia **collapsed**, killing 51 workers. The contractor was using a slip-formed construction process involving a multilevel scaffold that raises itself up the wall by its own power after anchoring into the hardened concrete of the previous day's work. According to the investigation …, the accident … one of the key factors … was "**a failure to make field tests to be sure that the concrete had cured sufficiently before the support forms were moved.**"

- *Eugene Kennedy, Sunday Examiner and Chronicle, December 3 1978*



4.6 Fresh and Early-age Properties

江西电厂坍塌事故

2016年11月24日7时40分许

意外发生在施工的最后一个步骤，当时来自河北的工人们正拆除冷凝塔外围的木制脚手架，但是尚未干透的混凝土开始脱落，最后坍塌。”

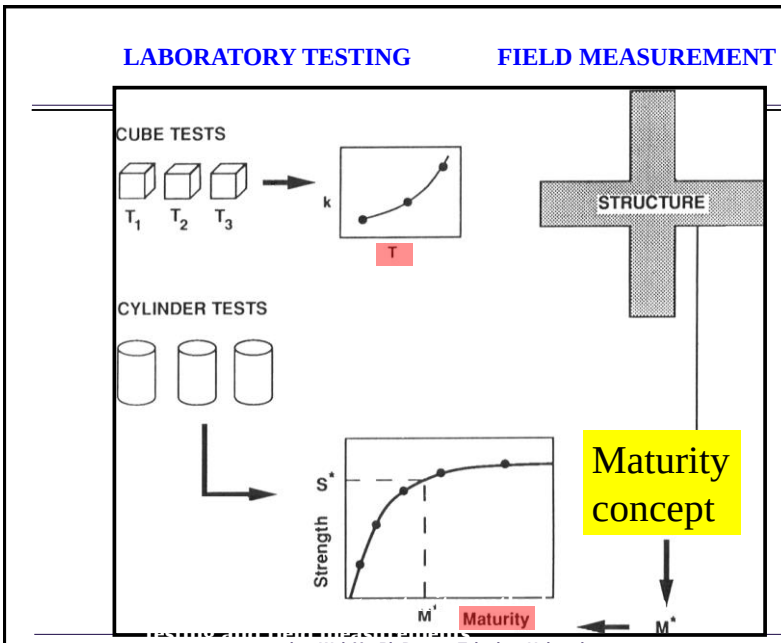
施工平台倒塌是罕见的。

工人从70多米高平台坠落

当时上下施工平台用的电梯，一起坍塌了。除了地面层的工友，在上面的人全部坠落，被钢筋等材料压在下面。”

并不是冷却塔发生了坍塌，而是在建冷却塔时搭建的一个非常大的施工平台发生了坍塌。





4.6 Fresh and Early-age Properties

Maturity Concepts

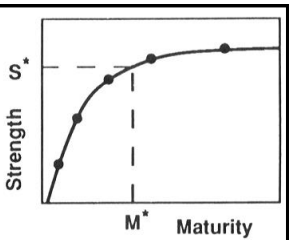
The maturity concept is based on the principle that strength gain in concrete is a function of curing time and temperature.

$$M = \sum (T - T_0)\Delta t$$

where
M = maturity
Σ = summation
T = concrete temperature, degrees Celsius
T_0 = datum temperature (基准温度).
Δt = duration of curing at temperature C, usually in hours

4.6 Fresh and Early-age Properti

Maturity Concepts



ACI 306 -88

1	2	3	4	5	6	7	8	9	10
Date	Elapsed time *h*, hr	Temperature in structure, F	Average temperature in structure F	Average temperature in structure C	Col. 5 + 5 C, C	Time interval *h*, hr	Col. 6x Col. 7 C hr	Maturity Σ Col. 8, C hr	Corresponding compressive strength psi
Sept. 1	0	50							
	12	50	50	10	15	12	180	180	-
2	24	50	50	10	15	12	180	360	-
	30	46	48	8.9	13.9	6	83	443	400
3	48	48	47	8.3	13.3	18	240	683	1080
	60	46	47	8.3	13.3	12	160	843	1400
4	72	44	45	7.2	12.2	12	146	989	1600
8	168	42	43	6.1	11.1	96	1065	2054	2600
11	240	42	42	5.6	10.6	72	763	2817	3100
14	312	42	42	5.6	10.6	72	763	3580	3400

4.6 Fresh and Early-age Properties

Maturity concept Equivalent age (Arrhenius Equations)

This function presents maturity in terms of equivalent age of curing at a specified temperature:

$$t_{eq} = \int_0^t \exp\left(\frac{E_a}{R}\left(\frac{1}{T_r} - \frac{1}{T(t)}\right)\right)dt$$

$$t_{eq} = \sum_0^t e^{\frac{E_a}{R}\left(\frac{1}{T_r} - \frac{1}{T(t)}\right)}\Delta t$$



t_{eq} : the equivalent age at the reference temperature

E_a : apparent activation energy, J/mol

$T(t)$: average concrete temperature (kelvin), during the time interval

T_r : absolute reference temperature (kelvin)=273+°C

Δt : time interval (hours) R : gas constant 8.314 J/mol-K,

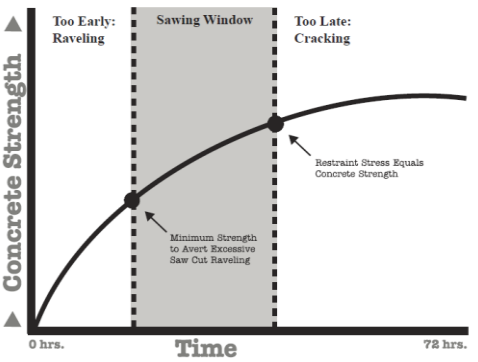
4.6 Fresh and Early-age Properties

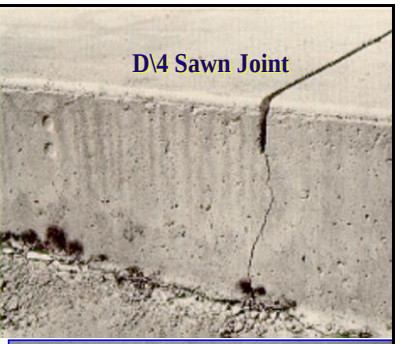
Maturity concept

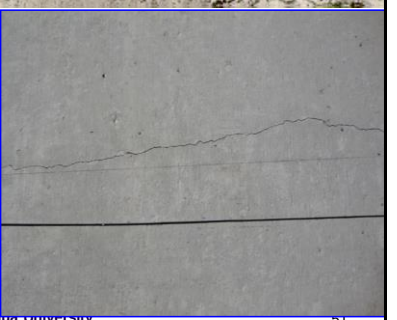


4.6 Fresh and Early-age P

Sawcuting timing and Depth







Summary

1. Mix Proportioning

Determination coarse aggregate and fine aggregate content

Determination of mixing water

Trial batch

2. Early-age Properties

Measurement of workability

Bleeding and segeragation

Plastic shrinkage



Questions?